Math
Spring Operational 2015

Grade 3
End of Year Released Items

VF888777

1. A brick path has 10 rows of 4 bricks. How many bricks are in the path?

Enter your answer in the box.

[ ]

VF656717

2. The movie theater in Vicky's town has 4 movie screening rooms. Each room has 58 seats on the left side and 32 seats on the right side.

**Part A**

Create an equation to find the total number of seats in the movie theater.

Enter your answers in the boxes.

( [ ] + [ ] ) × [ ] = ?

**Part B**

Round to the nearest ten the number of seats in the movie screening rooms to create a new equation. Solve the new equation.

Enter your answers in the boxes.

( [ ] + [ ] ) × [ ] = [ ]

VF906869

3. Fill in the numbers to complete each fact.

Enter your answers in the boxes.

$\boxed{\phantom{000}} = 7 \times 8$

$\boxed{\phantom{000}} = 63 \div 9$

$\boxed{\phantom{000}} = 40 \div 5$

$\boxed{\phantom{000}} = 7 \times 7$

$\boxed{\phantom{000}} = 48 \div 6$

M01071

4. A patio is in the shape of a rectangle with a width of 8 feet and a length of 9 feet. What is the area?

Enter your answer in the box.

$\boxed{\phantom{00}}$ square feet

VF564965

5. Drag and drop each fraction into the box labeled with an equivalent fraction.

| | | | |
|---|---|---|---|
| $\frac{1}{4}$ | $\frac{3}{6}$ | $\frac{4}{6}$ | $\frac{6}{8}$ |

| | | | |
|---|---|---|---|
| $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{2}{8}$ |

M02369

6. Which statements are true?

Select the **three** correct answers.

- [ ] A. $6 \times 6 = 9 \times 4$
- [ ] B. $8 \times 8 = 9 \times 6$
- [ ] C. $9 \times 8 = 7 \times 6$
- [ ] D. $81 \div 9 = 72 \div 8$
- [ ] E. $48 \div 6 = 64 \div 8$

VF525281

7. Complete this number sentence.

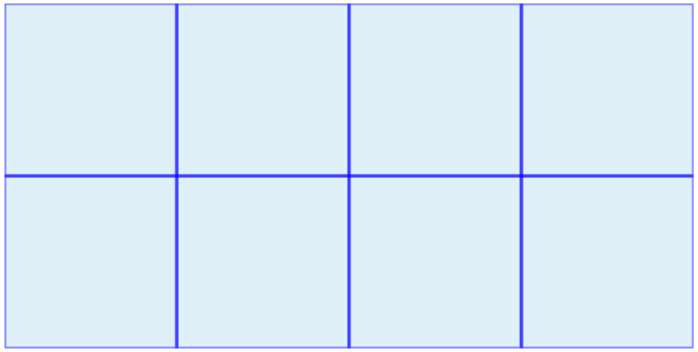
Enter your answer in the box.

$3 \times 90 =$ ☐

VF885478

8. Select sections of the model to shade a fraction equivalent to $\frac{3}{4}$ .

Select a section to shade it. Select a shaded section to remove the shading.

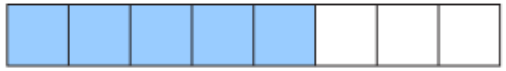


M01188

9. A model is divided into 8 equal parts. Which model shows the correct shading of $\frac{5}{8}$?

A.

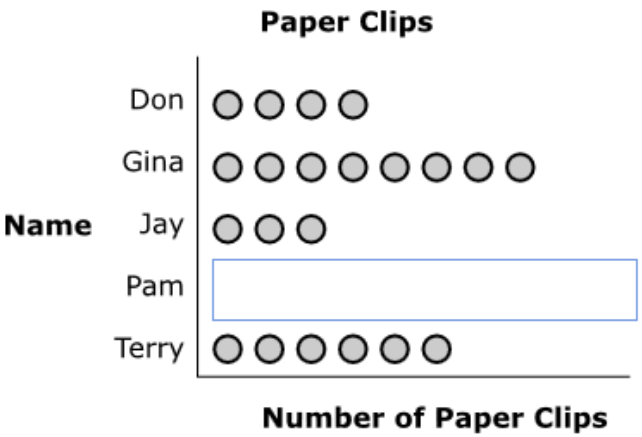
B.

C. 

D.

10. The picture graph shows the number of paper clips five friends each used to make paper clip chains. Pam's information is missing. Pam used 25 paper clips.

Complete the picture graph to show the number of paper clips Pam used. Drag and drop the circle into the picture graph. The circle may be used once, more than once, or not at all.




| KEY |
| --- |
| = 5 paper clips |

VF822882

11. Ava uses stickers to decorate picture frames. She has a total of 60 stickers. She uses 10 stickers on each picture frame.

When she uses all of her stickers, how many picture frames does Ava decorate?

Enter your answer in the box.

[ ] picture frames

VH000905

12. Several pizzas of the same size were left over at a party.

Select the **three** comparisons of the leftover pizza that are correct.

- [ ] A. $\frac{2}{8}$ of a pizza < $\frac{3}{8}$ of a pizza
- [ ] B. $\frac{3}{4}$ of a pizza < $\frac{3}{8}$ of a pizza
- [ ] C. $\frac{1}{3}$ of a pizza > $\frac{1}{4}$ of a pizza
- [ ] D. $\frac{3}{4}$ of a pizza = $\frac{2}{4}$ of a pizza
- [ ] E. $\frac{3}{6}$ of a pizza > $\frac{3}{8}$ of a pizza

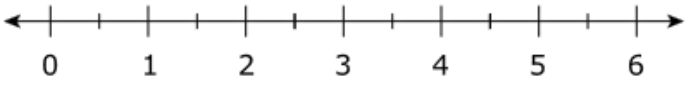
VF442827

13. A boat ride across a lake is 49 minutes each way. How many minutes does the boat ride take to go across the lake and back?

- A. 49
- B. 51
- C. 54
- D. 98

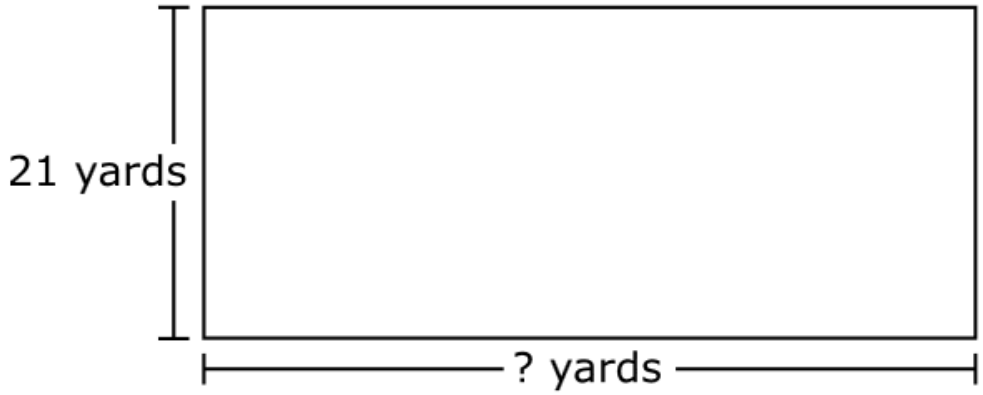
VF563153

14. Select the point that represents $\frac{5}{2}$ on the number line.



15. **Part A**

The rectangular garden at River Valley School is represented in the figure. The perimeter of the garden is 122 yards.




What is the missing side length in the figure?

Enter your answer in the box.

[ ] yards

**Part B**

River Valley School builds a new rectangular garden with the same perimeter, but the side lengths are different.

Which of these could be the side lengths of the new garden?

- A. 15 yards by 48 yards
- B. 14 yards by 47 yards
- C. 13 yards by 57 yards
- D. 14 yards by 58 yards

M00887

16. Which expressions are equal to the given expression?

$$157 + 748$$

Select the **three** correct answers.

- ☐ A. $150 + 755$
- ☐ B. $200 + 700$
- ☐ C. $90 + 5$
- ☐ D. $900 + 5$
- ☐ E. $(160 + 750) - (3 + 2)$
- ☐ F. $(9 + 100) + (0 + 10) + (5 + 1)$

VH011663

17. Select the boxes in the table to show whether each equation is true or false.

| | $72 \div 9 = 8$ | $5 \times 8 = 45$ | $36 \div 4 = 9$ | $42 \div 7 = 6$ | $9 \times 6 = 56$ |
|---|---|---|---|---|---|
| True | ○ | ○ | ○ | ○ | ○ |
| False | ○ | ○ | ○ | ○ | ○ |

M02022

18. Maya's rectangular rug has a perimeter of 16 feet. The length of the rug is 5 feet. What is the width of the rug?

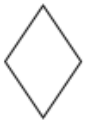
- A. 3 feet
- B. 9 feet
- C. 11 feet
- D. 13 feet

M02035

19. Which problem can be solved using the expression $3 \times 4$?

- A. A house has 3 rooms on the first floor and 4 rooms on the second floor. How many total rooms does the house have?
- B. A group of 4 friends share 3 large pizzas. How much pizza does each friend get?
- C. A shopping center has 3 floors, and each floor has 4 stores. How many total stores does the shopping center have?
- D. A group of friends spend $4 on French fries and $3 on drinks. How much do they spend on food and drinks?

20. For each shape pictured in the table, select the box for each correct shape name. You may select more than one box for each shape. If the shape is not a rectangle, rhombus, or quadrilateral, select the box for None of These.

| Shape | Rectangle | Rhombus | Quadrilateral | None of These |
|---|---|---|---|---|
| | ☐ | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ | ☐ |
|  | ☐ | ☐ | ☐ | ☐ |

VH011893

21. Which equations are true?

Select the **two** correct equations.

- [ ] A. $9 \times 9 = 81$
- [ ] B. $56 \div 7 = 9$
- [ ] C. $6 \times 8 = 48$
- [ ] D. $28 \div 4 = 8$
- [ ] E. $3 \times 9 = 33$

M01877

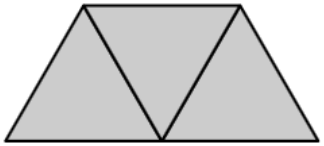
22. A store has two different-sized fish tanks. One fish tank holds 218 liters of water, and the other fish tank holds 145 liters of water. What is the total number of liters for the two fish tanks?

- A. 133 liters
- B. 173 liters
- C. 353 liters
- D. 363 liters

23. Which shapes are divided into thirds?

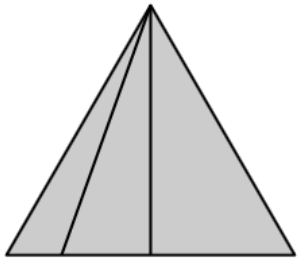
Select the **three** correct answers.

☐ A.

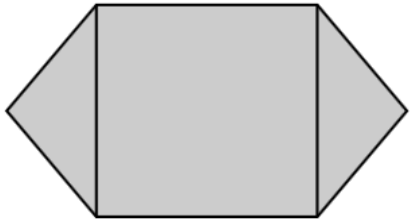


☐ B.

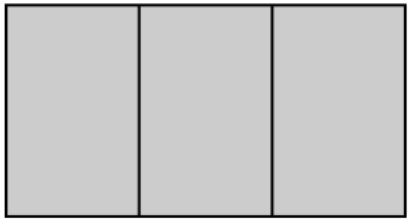
☐ C.



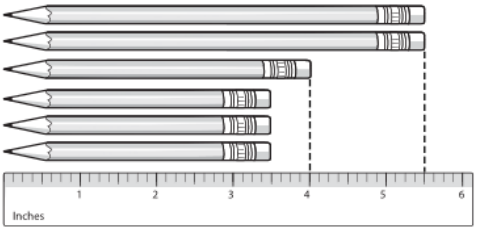
☐ D.



☐ E.

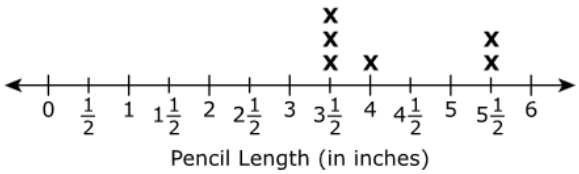


24. Brian measured the lengths, in inches, of the pencils in his desk as shown.

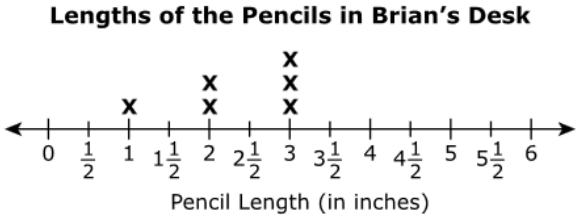



Which of these line plots shows the length of each pencil plotted correctly?

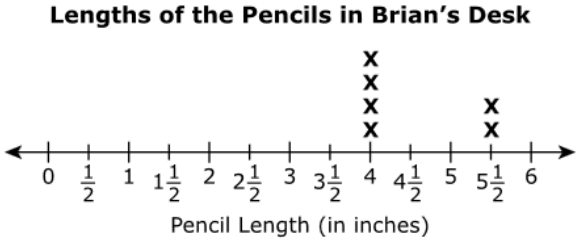
A.



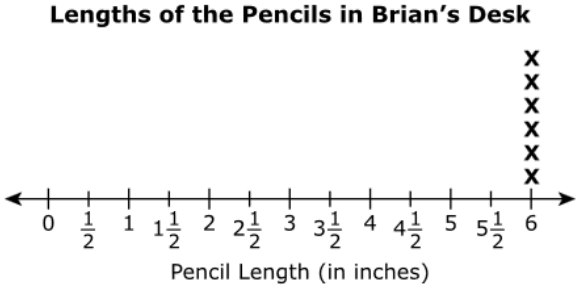

B.




C.




D.



VH003125

25. Mr. Caden builds a fence around his rectangular backyard that is 8 meters long and 7 meters wide.

What is the perimeter of the backyard?

Enter your answer in the box.

[ ] meters

M02037

26. For a field trip, 48 students are put into groups with 6 students in each group. Which expression can be used to show the total number of groups formed?

- A. $48 \div 6$
- B. $6 \div 48$
- C. $6 \times 48$
- D. $48 + 6$

VF906806

27. Correctly complete the equation.

Enter your answer in the box.

[ ] $\div 5 = 8$

M01400

28. What is $908 - 412$?

- A. 416
- B. 496
- C. 504
- D. 596

VH012290

29. Drag and drop a number to correctly complete each equation. Not all numbers will be used.

630 | 24 | 80 | 90 | 300 | 900 | 30 | 800 | 240 | 63

$80 \times 3 =$ ☐

$30 \times 3 =$ ☐

$50 \times 6 =$ ☐

$40 \times 2 =$ ☐

$90 \times 7 =$ ☐

30. **Part A**

Freda buys horse food in 20-kilogram bags. Her horse eats 8 bags of horse food per month.

How much horse food does Freda's horse eat in one month?

Enter your answer in the box.

[ ] kilograms

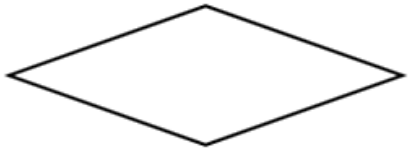
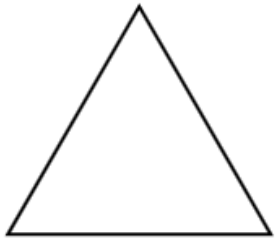
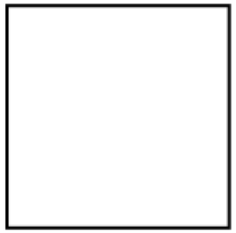
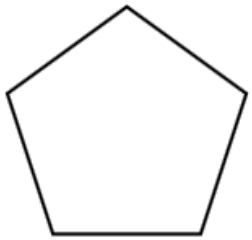
**Part B**

Freda's horse has a mass of 782 kilograms. Kurt's pony has a mass of 359 kilograms. How much more mass does Freda's horse have than Kurt's pony?

Enter your answer in the box.

[ ] kilograms

31. Which **three** shapes are quadrilaterals?

- ☐ A.
- ☐ B. 
- ☐ C. 
- ☐ D. 
- ☐ E. 

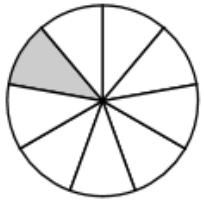
32. Janell has 8 photos on one page of an album. She has 4 photos on each of the other 9 pages. How many photos are in Janell's album?

Enter your answer in the box.

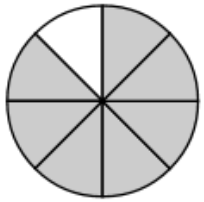
[ ] photos

33. Which circle is divided into 8 equal parts and has $\frac{1}{8}$ **shaded**?

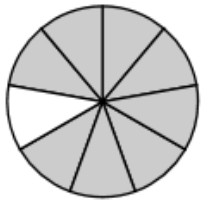
A. **Circle**



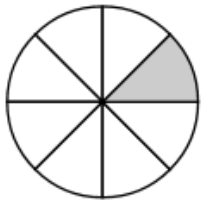
B. **Circle**



C. **Circle**



D. **Circle**



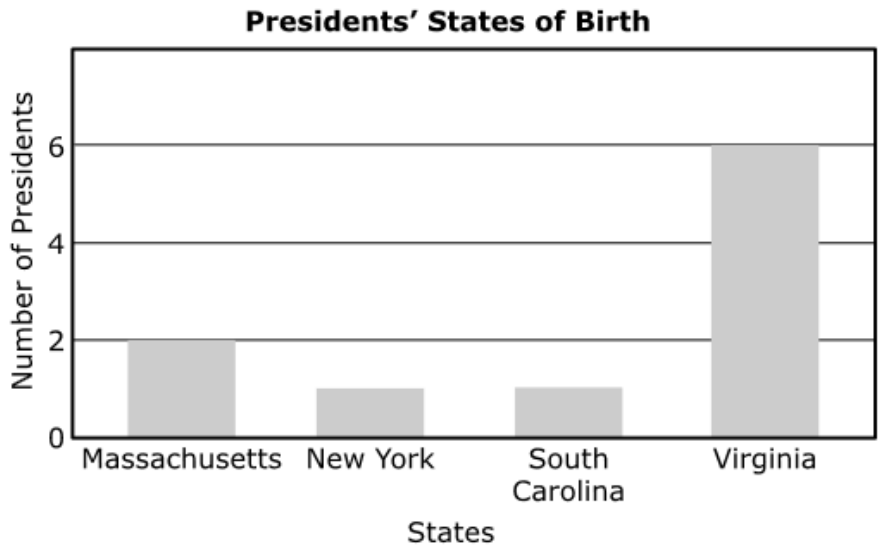
VF556343

34. Enter your answer in the box.

$263 + 449 =$ _____

0530-M00067

35. The first 10 presidents of the United States were born in four states. The bar graph shows the number of presidents born in each state.




**Part A**

How many more presidents were born in Virginia than in New York?

Enter your answer in the box.

_____

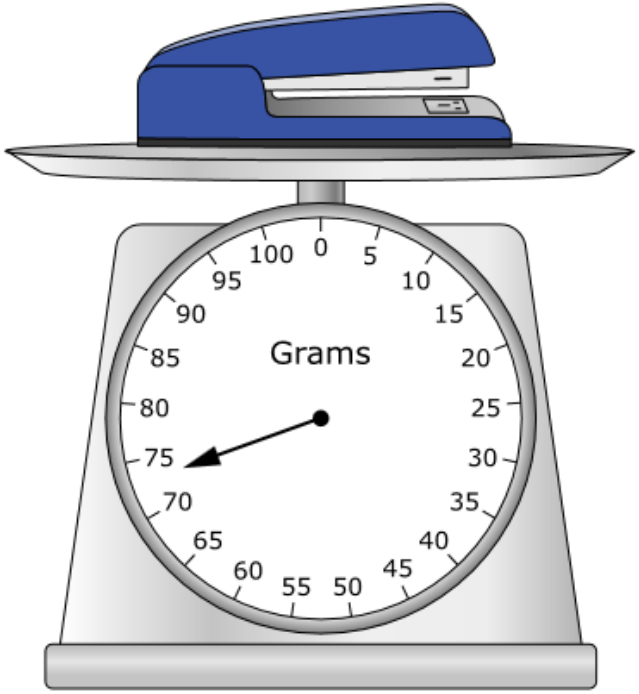
**Part B**

What is the difference between the number of presidents born in Massachusetts and the number of presidents born in New York and South Carolina together?

Enter your answer in the box.

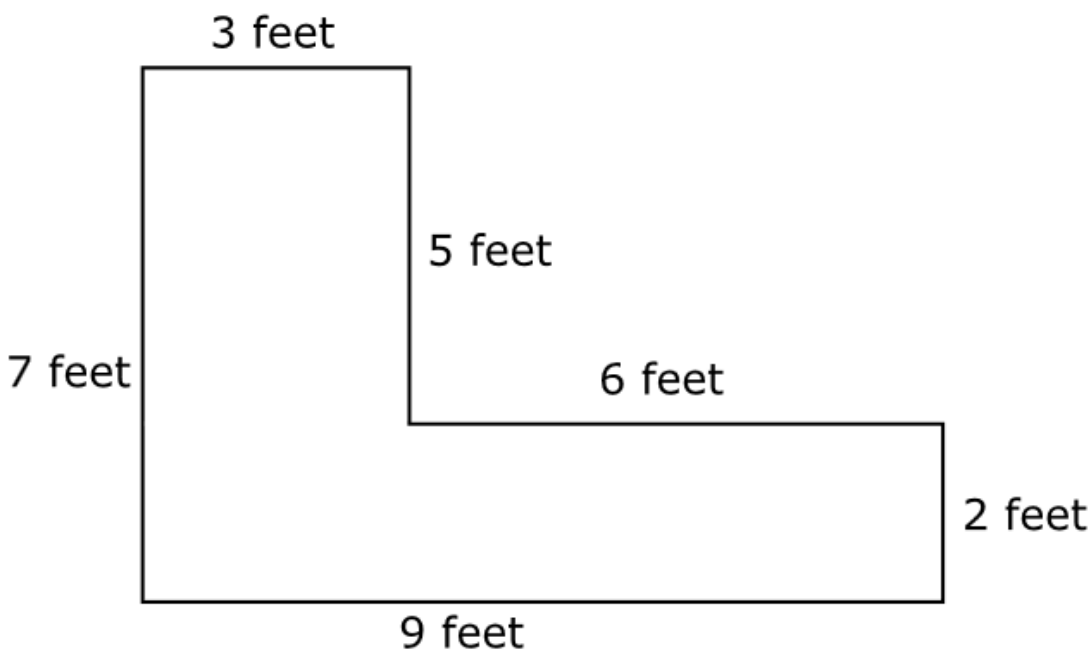
_____

36. Which is closest to the mass of the stapler?




- A. 15 grams
- B. 25 grams
- C. 65 grams
- D. 75 grams

37. Rex's garden is made from two rectangles as shown.




What is the area of Rex's garden?

Enter your answer in the box.

[ ] square feet

38. Select the **two** fractions that are equivalent to 1.

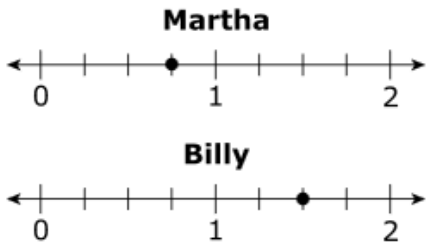
- [ ] A. $\frac{3}{1}$
- [ ] B. $\frac{2}{2}$
- [ ] C. $\frac{4}{3}$
- [ ] D. $\frac{6}{6}$
- [ ] E. $\frac{1}{8}$

39. Martha used $\frac{3}{4}$ cup of sauce, and Billy used $\frac{6}{4}$ cups of sauce. They were each supposed to use 1 cup of sauce.

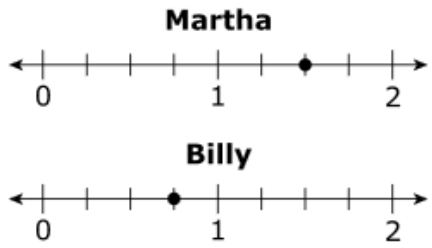
**Part A**

Which set of number lines show the amount of sauce, in cups, that Martha and Billy used?

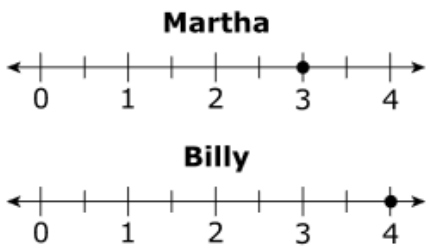
A.



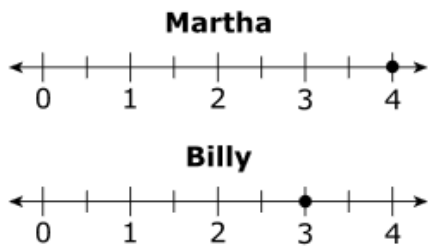

B.




C.




D.




**Part B**

Which statement is correct?

A. Billy used less sauce than he was supposed to use.

B. Martha used more sauce than she was supposed to use.

C. Billy was closer than Martha to using 1 cup of sauce.

D. Martha was closer than Billy to using 1 cup of sauce.